Do Now · retrieval

TCP/IP stack, protocols and DNS

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which device typically forwards packets between different IP networks?

Q2

What does an edge represent in a network graph?

Q3

How many bits make one byte?

Q4

A sender and receiver are on different networks. Can a switch alone normally route IP packets between them?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Start with a hostname — The browser uses DNS (an application-layer protocol) to resolve school.example to an IP address. A cached result may avoid further queries.

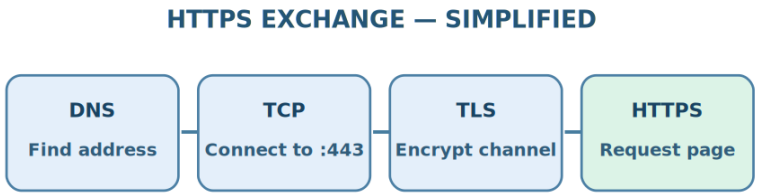
Core idea 2

Understand four layers — Application supports services such as HTTP(S), DNS and SMTP; transport includes TCP and UDP; internet uses IP; link transfers frames on the local connection.

Core idea 3

Build an HTTPS connection — A browser normally establishes TCP to port 443, negotiates TLS, then exchanges encrypted HTTP requests and responses. DNS lookup happens first if the IP is unknown.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

application layer — Protocols offering services to applications, including DNS and HTTP.

transport layer — End-to-end transport protocols such as TCP and UDP.

internet layer — Network addressing and forwarding using IP.

link layer — Local transfer of frames over media such as Ethernet or Wi-Fi.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A user opens https://library.example. Explain what DNS, TCP and HTTPS each do.

Follow the worked method

1. Name resolution — The resolver returns the IP address for library.example (or uses a cached answer).

2. Transport connection — TCP establishes a connection to the server, typically destination port 443.

3. Application exchange — TLS secures the connection; HTTPS carries the page request and response.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Identify the application
SMTP is an application-layer protocol for transferring outgoing email.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Which TCP/IP layer contains DNS, and what does DNS do? [2]

Question 2

Place TCP, IP, and HTTP into the correct layers and explain one role of each. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A pupil says DNS encrypts webpage traffic and TCP chooses the Wi-Fi channel. Correct both errors, identifying the relevant protocols/layers. [4]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

DNS and HTTPS are both application-layer protocols in the TCP/IP model.

Use key vocabulary from memory

1. Explain application layer in your own words.

2. Explain transport layer in your own words.

3. Explain internet layer in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.